



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

. Ravi Kumar Reddy(192424304)

Lab Programs

1.student_average.py

Code:

```
import numpy as np

student_scores = np.loadtxt(
    "Students_data.csv",
    delimiter=",",
    skiprows=1,
    usecols=(1, 2, 3, 4)
)

subjects = ["Math", "Science", "English", "History"]
average_scores = np.mean(student_scores, axis=0)
highest_subject = subjects[np.argmax(average_scores)]

print("Average Scores:", np.round(average_scores, 2))
print("Subject with Highest Average Score:", highest_subject)
```

Output:

```
Average Scores: [82.  88.  83.2 84.8]
Subject with Highest Average Score: Science
>>>
```

2.Past_month_price_average.py

Code:

```
import pandas as pd
df = pd.read_csv("Sales_data.csv")
average_price = df[df["Month_sales"] == "November 2023"]["Price"].mean()
print("Average Price in November 2023:", round(average_price, 2))
```

Output:

```
>>> | Average Price in November 2023: 25000.0
```

3. house_average_price.py

Code:

```
import numpy as np
house_data = np.loadtxt(
    "House_data.csv",
    delimiter=",",
    skiprows=1
)
houses_more_than_4 = house_data[house_data[:, 1] > 4]
average_price = np.mean(houses_more_than_4[:, 3])
print("Average Price of Houses with more than 4 Bedrooms:", round(average_price, 2))
```

Output:

```
>>> | Average Price of Houses with more than 4 Bedrooms: 15500000.0
```

4. quarterly_sales_numpy.py

Code:

File Edit Format Run Options Window Help

```
import numpy as np

sales_data = np.genfromtxt(
    "Quarterly_sales_data.csv",
    delimiter=",",
    skip_header=1,
    dtype=str
)

months = sales_data[:, 1]
sales = sales_data[:, 4].astype(float)

Q1 = 0.0
Q4 = 0.0

for i in range(len(months)):
    if months[i] in ["January", "February", "March"]:
        Q1 += sales[i]
    elif months[i] in ["October", "November", "December"]:
        Q4 += sales[i]

total_sales_year = np.sum(sales)

percentage_increase = ((Q4 - Q1) / Q1) * 100

print("Total Sales for the Year:", round(total_sales_year, 2))
print("Percentage Increase from Q1 to Q4:", round(percentage_increase, 2), "%")
```

Output:

```
Total Sales for the Year: 590000.0
Percentage Increase from Q1 to Q4: 50.0 %
>>>
```

5. fuel_efficiency_analysis.py

Code:

File Edit Format Run Options Window Help

```
import numpy as np

fuel_data = np.genfromtxt(
    "Fuel_data.csv",
    delimiter=",",
    skip_header=1,
    dtype=str
)

fuel_efficiency = fuel_data[:, 7].astype(float)
make = fuel_data[:, 8]

average_efficiency = np.mean(fuel_efficiency)

mazda_eff = fuel_efficiency[make == "mazda"]
audi_eff = fuel_efficiency[make == "audi"]

mazda_avg = np.mean(mazda_eff)
audi_avg = np.mean(audi_eff)

percentage_improvement = ((mazda_avg - audi_avg) / audi_avg) * 100

print("Average Fuel Efficiency of all cars:", round(average_efficiency, 2), "MPG")
print("Average Fuel Efficiency of Mazda:", round(mazda_avg, 2), "MPG")
print("Average Fuel Efficiency of Audi:", round(audi_avg, 2), "MPG")
print("Percentage Improvement from Mazda to Audi:", round(percentage_improvement, 2), "%")
```

Output:

```
Average Fuel Efficiency of all cars: 28.67 MPG
Average Fuel Efficiency of Mazda: 31.0 MPG
Average Fuel Efficiency of Audi: 27.0 MPG
Percentage Improvement from Mazda to Audi: 14.81 %
>>>
```

6. customer_purchase.py

Code:

```
File Edit Format Run Options Window Help
import pandas as pd

df = pd.read_csv("grocerystore.csv")

total_sales = df["Sales"].sum()

print("Total sales for the grocery store:", round(total_sales, 2))

discount = 0.10
tax = 0.18

price_after_discount = total_sales - (total_sales * discount)

price_after_tax = price_after_discount + (price_after_discount * tax)

print("Final amount after applying discount and tax:", round(price_after_tax, 2))
```

Output:

```
Total sales for the grocery store: 3850
Final amount after applying discount and tax: 4088.7
>>>
```

7.customer_order_analysis.py

Code:

```

File Edit Format Run Options Window Help
import pandas as pd

df = pd.read_csv("order_data.csv")

total_orders = df.groupby("Full Name")["Order Count"].sum()

avg_order_per_product = df.groupby("Items")["Order Total"].mean()

earliest_order = df["Order Date"].min()

latest_order = df["Order Date"].max()

print("Total number of orders made by each customer:")
print(total_orders)

print("\nAverage order total for each product:")
print(avg_order_per_product)

print("\nEarliest order date:", earliest_order)

print("Latest order date:", latest_order)

```

Output:

```

Total number of orders made by each customer:
Full Name
Anjali      3
Priya       3
Rahul       5
Name: Order Count, dtype: int64

Average order total for each product:
Items
Headphones      4000.0
Laptop          75000.0
Mobile          40000.0
Name: Order Total, dtype: float64

Earliest order date: 2023-01-10
Latest order date: 2023-05-18
>>>

```

8. five_most_sold_products.py

Code:

```

File Edit Format Run Options Window Help
import pandas as pd

df = pd.read_csv("Electronics_data.csv")

top_5_products = df.groupby("Product Name")["Price"].sum().nlargest(5)

print("Five Most Sold Products:")
print(top_5_products)

```

Output:

```

Five Most Sold Products:
Product Name
Laptop      165000
TV          62000
Mobile      45000
Refrigerator 45000
AC          40000
Name: Price, dtype: int64
>>>

```

